chapter seven

SAMPLING DISTRIBUTIONS

I know of scarcely anything
so apt to impress the imagination as the wonderful form
of cosmic order expressed by the
"Law of Frequency of Error" (the normal distribution).
– Sir Francis Galton

INTRODUCTION

In chapter 6 we were introduced to a continuous random variable. We then looked at the standard normal distribution and used the z score to find the area under the **standard normal curve**. We were also able to convert a normal distribution involving a continuous random variable x into a standard normal distribution. We then used the z value to find an area under the normal distribution curve.

In this chapter we extend the idea of the probability distribution of a continuous random variable x to the probability distributions of the sample statistics $\bar{x}$ and $\hat{p}$. It is important to note that a sample statistic is a random variable and we refer to the probability distribution of the sample statistic as the **sampling distribution**. Using what is called the **Central Limit Theorem**, we will be able to use the z score to transform the sampling distribution of $\bar{x}$ and $\hat{p}$ into standard normal distributions. The Central Limit Theorem will tell us how to find the **mean and variance of the sampling distribution**. We will then be able to find the probabilities for values of $\bar{x}$ or $\hat{p}$.

As an example, suppose we were interested in the distribution of car weights and want to find the probability that the average weight of 100 randomly selected cars is greater than 2,000 pounds. We need to know a value for the population mean, μ. For the present we will be given the value of the population mean. But later on this value will be obtained from a hypothesis or belief that someone has about the distribution of car weights. Once we have the value of μ, we use the Central Limit Theorem to find the mean and standard deviation of the sampling distribution of $\bar{x}$. We then find the z score for $\bar{x}$ and find the desired probability by finding the area under the right tail of the standard normal curve.

The Central Limit Theorem states that for $n \geq 30$, the sampling distribution of the sample mean is approximately normal irrespective of the shape of the population distribution, and

$$\mu_{\bar{x}} = \mu \text{ and } \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}.$$

The beauty of the Central Limit Theorem is that no matter what the distribution of the continuous variable x, the sampling distribution of $\bar{x}$ will be approximately normal as long as $n \geq 30$.

In a similar manner if np and nq are both greater than 5, the sampling distribution of a sample proportion, $\hat{p}$, is approximately normal, and

$$\mu_{\hat{p}} = p \text{ and } \sigma_{\hat{p}} = \sqrt{\frac{pq}{n}}$$

The steps outlined in the above example on the distribution of car weights become the essence of what we will call hypothesis testing in the chapters that follow. It will be very important that you understand what the Central Limit Theorem states.

CHAPTER 7 HOT SPOTS

1. **The Difference Between $P(x \geq 65)$ And $P(\bar{x} \geq 65)$.**
   Starts on **page 7–4.** Problems on **pages 7–8, 7–9, 7–10.**
   Mann 7.5

If you find other HOT SPOTS, write them down and use them as a focus of your discussions in the study group. Or you can use the HOT SPOT as the topic for a help session with your professor.

✷
Mann
Section 7.5

Hot Spot #1 – The Difference Between $P(x \geq 65)$ And $P(\bar{x} \geq 65)$

One area of difficulty in this chapter is working with the Central Limit Theorem. It is fairly easy to know that the following two statements are part of the CLT:

$$\mu_x = \mu_{\bar{x}} \qquad\qquad \sigma_{\bar{x}} = \frac{\sigma_x}{\sqrt{n}}$$

However, the real problem is to understand the effect these two statements have on a problem we are doing. As an example, we will look at the two questions from our example using the continuous random variable x = the speed of a car where:

$$\mu = 55 \text{ m.p.h. and } \sigma = 5 \text{ m.p.h.}$$

The first question involves finding the probability that a single speed is greater than 65. We write the first question as

$$P(x \geq 65)$$

The second question involves finding the probability that the average speed from a sample of n cars is more than 65. We write the second question as

$$P(\bar{x} \geq 65)$$

In order to understand the difference in these two questions, we need to first look at the distribution of speeds. Figure 7–1 shows the bell-shaped curve. The area shaded represents the 95% of the speeds between 45 m.p.h. and 65 m.p.h. (using the Empirical Rule).

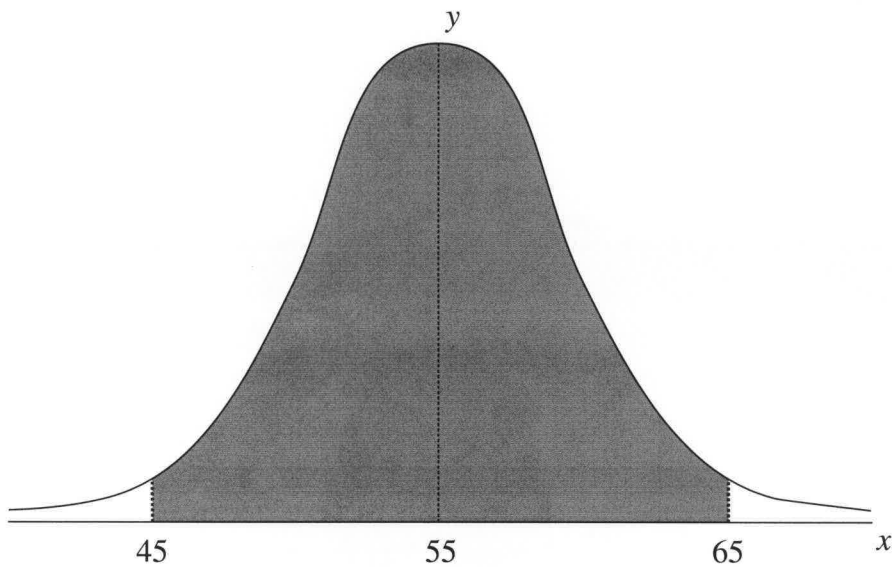

Figure 7–1: 95% Of Speeds Between 45 M.P.H. and 65 M.P.H.

If we take a sample of speeds from this distribution, it will be unusual to get speeds greater than 65 m.p.h.. In fact, we will only get speeds greater than 65 m.p.h. approximately 2.5% of the time. We can now answer the first question as follows:

$$P(x \geq 65) = 0.025$$

Therefore, 97.5% of the time we would get speeds less than 65 m.p.h.. As a result, the average speed of any sample we take should usually be less than 65 m.p.h.. The difficult part of the question is to quantify the "usually less". This is where we make use of the Central Limit Theorem. The CLT tells us that the standard deviation of the distribution of the $\bar{x}$'s that we get from sampling has a value of 5 m.p.h. divided by the square root of the sample size. If we take a sample of 25 speeds, the standard deviation of the $\bar{x}$'s (sample means) will be $\frac{5}{\sqrt{25}}$.

$$\sigma_{\bar{x}} = 1 \text{ m.p.h..}$$

The bell-shaped curve for the distribution of $\bar{x}$ will be centered at $\mu = 55$, but the standard deviation will now be 1. The curve is shown below: (only the right half of the curve is shown to save space)

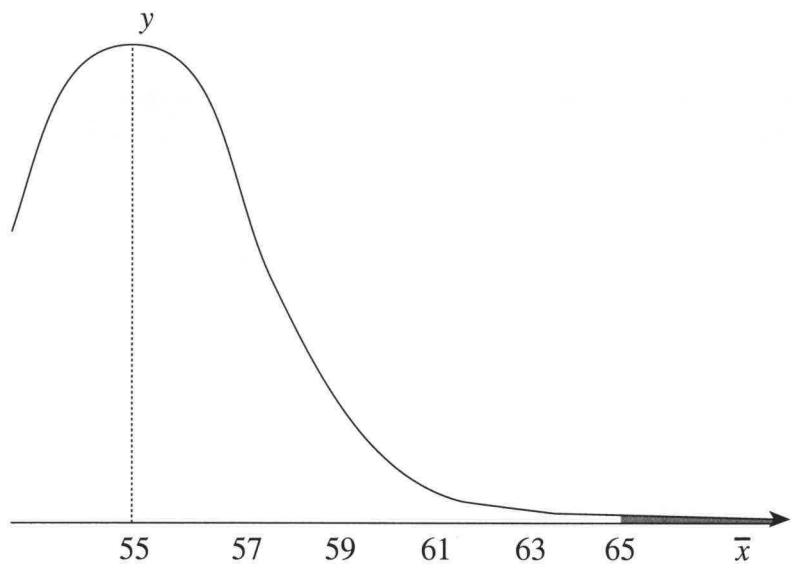

Figure 7–2: Bell Curve For Sampling Distribution Of Means

As you can see, there is very little of the area under the curve to the right of 65 m.p.h.. We are not able to use the Empirical Rule to find our answer to the second question $P(\bar{x} \geq 65)$, but we can find a z value and use either the z table or the HP-21S calculator to find our answer. The z value is found as follows:

$$z = \frac{65 - 55}{1} = 10$$

Unfortunately our z table only goes up to 3.49. The probability that z is greater than 3.49 is given below:

$$P(z \geq 3.49) = 1 - 0.9998 = 0.0002$$

Therefore the probability that z is greater than 10 must be smaller than the number we just found. If we use the HP-21S with the key that gives us the area under the normal curve, we find the exact answer of $P(z \geq 10) = 7.62 \times 10^{-24} = 0.0^{+}$.

We have found that although we might expect to occasionally find one speed selected from the population to be greater than 65, it would be extremely unusual to find a mean of 25 speeds selected from the population to be more than 65. Indeed, this result is the heart of what we do next in the chapter on hypothesis testing. Whenever we get a value that is rare, we consider it to be significant. If we do get such a rare value, we conclude that our original assumption (hypothesis) that μ was 55 m.p.h. is in error and our hypothesis should be rejected.

The two problems we just finished, finding

$$P(x \geq 65) \text{ and } P(\bar{x} \geq 65)$$

make it look like we are working with separate problems, but there is actually a very important common structure to the two problems. Both problems use the z score. We can write the z score in a form that will be very useful as we continue working in the next chapters. We can treat the z score using a template found in spread sheets like LOTUS 1-2-3. We simply fill in the box with the appropriate symbol.

The same z formula is used for several different z score calculations, with one substitution for each z score.

$$\boxed{z = \frac{[\,] - \mu_{[\,]}}{\sigma_{[\,]}} \text{ where the } [\,] \text{ can be } \begin{matrix} 1.\ x \\ 2.\ \bar{x} \end{matrix}}$$

We are actually using the same formula each time, with the box representing the statistic we are using. Later on we will increase the number of choices for the box to include a difference of sample means, a sample proportion, a difference of sample proportions, and a correlation coefficient. The denominator of the z score is called the standard error of the statistic we are using. When the sample statistic is $\bar{x}$, $\sigma_{\bar{x}}$ is referred to as the standard error of the mean. We often refer to $\sigma_{\bar{x}}$ simply as the standard error.

As we know from the Central Limit Theorem, the original population does not have to be normally distributed to use the theorem for a distribution of sample means. Furthermore, we do not need to know the value of σ if the sample size is considered large, $n \geq 30$. If the sample size is large we can use the value of s, the sample standard deviation, in place of the value of σ. As long as $n \geq 30$ we can write the z score as follows:

$$z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

At this point, the use of s for σ may not seem like an important issue, but if $n < 30$ we will have to adjust the formula because s will be an underestimate of σ. We will talk more about this in the chapter that covers small samples using what is called the t distribution.

Hot Spot #1 Sample Problem and Solution: Given a normal distribution of cholesterol readings with $\mu = 200$ and $\sigma = 20$ find the following:

A. $P(x \leq 180)$ B. $P(\bar{x} \leq 180)$ with $n = 100$

Solution:

(A.) Step 1. Graph the bell-shaped curve.

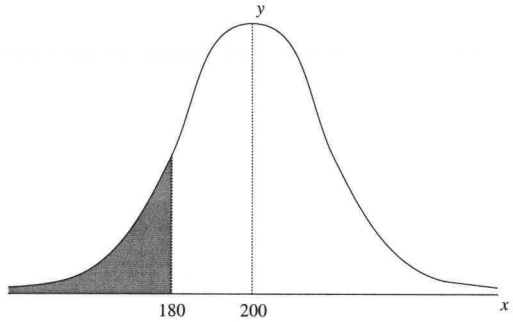

Figure 7-3: Cholesterol Readings

Step 2. Find the z score for $x = 180$.

$$z = \frac{180 - 200}{20} = -1$$

Step 3. $P(x \le 180) = P(z \le -1) \ = 0.1587$

$P(x \le 180) = 0.1587$

(B.) Step 1. Graph the bell-shaped curve.

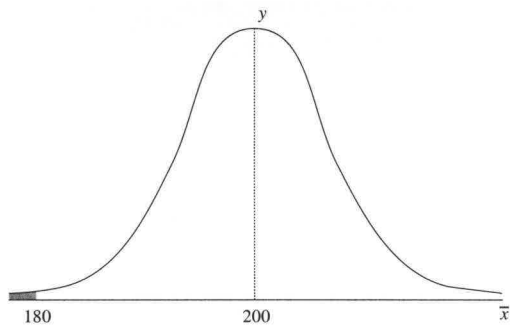

Figure 7–4: Average Cholesterol Readings

Step 2. Find the z score for $\bar{x} = 180$.

$$z = \frac{180 - 200}{\frac{20}{\sqrt{100}}} = -10$$

Step 3. $P(\bar{x} \le 180) = P(z \le -10) = 0.0^{+}$

Sample Problem and Answer: Given that x has a normal distribution with $\mu = 70$ and $\sigma = 10$ find $P(\bar{x} \ge 73)$ when $n = 100$.

Answer: $P(\bar{x} \ge 73) = P(z \ge 3) = 1 - 0.9987 = 0.0013$

Sample Problem and Answer: Given that x has a normal distribution with $\mu = 110$ and $\sigma = 5$ find $P(x \le 100)$.

Answer: $P(x \le 100) = P(z \le -2) = 0.0228$.

CHAPTER 7 DISCUSSION QUESTIONS

These questions may be used in your study group or simply as topics for individual reflection. Whichever you do, take time to explain verbally each topic to insure your own understanding. Since these questions are intended as topics for discussion, answers to these questions are not provided. If you find that you are not comfortable with either your answers or that your group has difficulty with the topic, take time to meet with your professor to get help.

1. What is the sampling error?

2. What does the Central Limit Theorem say about the mean and standard deviation of the random variables $\bar{x}$ and $\hat{p}$?

3. What happens to $\sigma_{\bar{x}}$ and $\sigma_{\hat{p}}$ as n gets larger?

4. How does one control non-sampling errors ?

5. What is an important difference between the population parameter and the random variable $\bar{x}$?

6. Relative to the use of the finite correction factor, when is a sample size considered to be small?

7. What is the finite population correction factor and when should it be used?

8. What is the shape of the sampling distribution of $\bar{x}$ if $n \geq 30$, and and $\hat{p}$ if $np>5$ and $nq>5$?

9. What are the z scores for $\bar{x}$ and $\hat{p}$?

10. What is the difference between a population distribution and a sampling distribution?